

Problem

Determine the Fourier transforms of the following functions: (a) gate function $g(t) = 10u(t - 1) - 10u(t - 2)$, (b) $12\delta(t - 2)$, (c) $15 \sin \omega_0 t$.

Step-by-step solution

Step 1 of 3

The expression for the Fourier transform of the function $f(t)$ is,

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

(a)

The gate function is,

$$g(t) = 10u(t - 1) - 10u(t - 2)$$

The Fourier transform of the gate function is,

$$\begin{aligned} G(\omega) &= \int_{-\infty}^{\infty} g(t) e^{-j\omega t} dt \\ &= \int_{-\infty}^{\infty} [10u(t - 1) - 10u(t - 2)] e^{-j\omega t} dt \\ &= \int_{-\infty}^{\infty} [10u(t - 1) e^{-j\omega(t-1)} e^{-j\omega} - 10u(t - 2) e^{-j\omega(t-2)} e^{-j2\omega}] dt \end{aligned}$$

Using shifting property and Fourier Transform of $u(t)$, $F[u(t)] = \frac{1}{j\omega}$.

$$\begin{aligned} G(\omega) &= \frac{10e^{-j\omega}}{j\omega} - \frac{10e^{-j2\omega}}{j\omega} \\ &= \frac{10(e^{-j\omega} - e^{-j2\omega})}{j\omega} \end{aligned}$$

Thus, the Fourier transform of the gate function $g(t)$ is $\boxed{\frac{10(e^{-j\omega} - e^{-j2\omega})}{j\omega}}$.

Step 2 of 3

(b)

Consider that the function is,

$$f(t) = 12\delta(t-2)$$

$\delta(t-2)$ is unit impulse function which is delayed by two units and has value only at $t = 2$.

The Fourier transform of the function is,

$$\begin{aligned} F(\omega) &= \int_{-\infty}^{\infty} 12\delta(t-2)e^{-j\omega t} dt \\ &= \int_{-\infty}^{\infty} 12\delta(t-2)e^{-j\omega(t-2)}e^{-j2\omega} dt \end{aligned}$$

Using shifting property of impulse,

$$\begin{aligned} F(\omega) &= 12e^{-j2\omega} \int_{-\infty}^{\infty} \delta(t-2)e^{-j\omega(t-2)} dt \\ &= 12e^{-j2\omega} (1) \\ &= 12e^{-j2\omega} \end{aligned}$$

Thus, the Fourier transform of the function is $12e^{-j2\omega}$.

Step 3 of 3

(c)

Consider that the function is,

$$f(t) = 15 \sin \omega_0 t$$

The Fourier transform of the function is,

$$\begin{aligned} F[15 \sin \omega_0 t] &= F\left[15 \frac{e^{j\omega_0 t} - e^{-j\omega_0 t}}{2j}\right] \\ &= \frac{15}{2j} F[e^{j\omega_0 t} - e^{-j\omega_0 t}] \\ &= \frac{15}{2j} \{F[e^{j\omega_0 t}] - F[e^{-j\omega_0 t}]\} \end{aligned}$$

Using the transform $F[e^{j\omega_0 t}] = 2\pi\delta(\omega - \omega_0)$,

$$\begin{aligned} F(\omega) &= \frac{15}{2j} [2\pi\delta(\omega - \omega_0) - 2\pi\delta(\omega + \omega_0)] \\ &= \frac{15}{j} [\pi\delta(\omega - \omega_0) - \pi\delta(\omega + \omega_0)] \\ &= -j15 [\pi\delta(\omega - \omega_0) - \pi\delta(\omega + \omega_0)] \\ &= j15 [\pi\delta(\omega + \omega_0) - \pi\delta(\omega - \omega_0)] \end{aligned}$$

Thus, the Fourier transform of the function is $j15 [\pi\delta(\omega + \omega_0) - \pi\delta(\omega - \omega_0)]$.